



The Bombay Salesian Society's Don Bosco Institute of Technology Mumbai
(An Autonomous Institute Affiliated to University of Mumbai)
Department of Electronics and Telecommunication Engineering
SET4PCC03 Digital VLSI Design Laboratory

Modelling code in the Software (Ngspice):

Before Scaling:

```
cmos inverter Neha - 18 |
VDD 1 0 DC 5V
VIN 3 0 DC 5V
VD 1 2 DC 0V
MP 4 3 2 2 PMOD W=20U L=1U
MN 4 3 0 0 NMOD W=10U L=1U
CL 4 0 10pF
.MODEL PMOD PMOS (KP=50U,VTO=-1V)
.MODEL NMOD NMOS (KP=100U,VTO=1V)
.DC VIN 0 5 0.05
.control
run
plot V(4) V(3)
plot deriv(V(4))
plot i(VD)*V(1)
.endc
.end
```

After Constant Voltage Scaling (CVS):

```
cmos inverter Neha - 18 (sf = 1.9) cvs|
VDD 1 0 DC 5V
VIN 3 0 DC 5V
VD 1 2 DC 0V
MP 4 3 2 2 PMOD W=10.52U L=0.5263U
MN 4 3 0 0 NMOD W=5.263U L=0.5263U
CL 4 0 2.77pF
.MODEL PMOD PMOS (KP=95U,VTO=-1V)
.MODEL NMOD NMOS (KP=190U,VTO=1V)
.DC VIN 0 5 0.05
.control
run
plot V(4) V(3)
plot deriv(V(4))
plot i(VD)*V(1)
.endc
.end
```



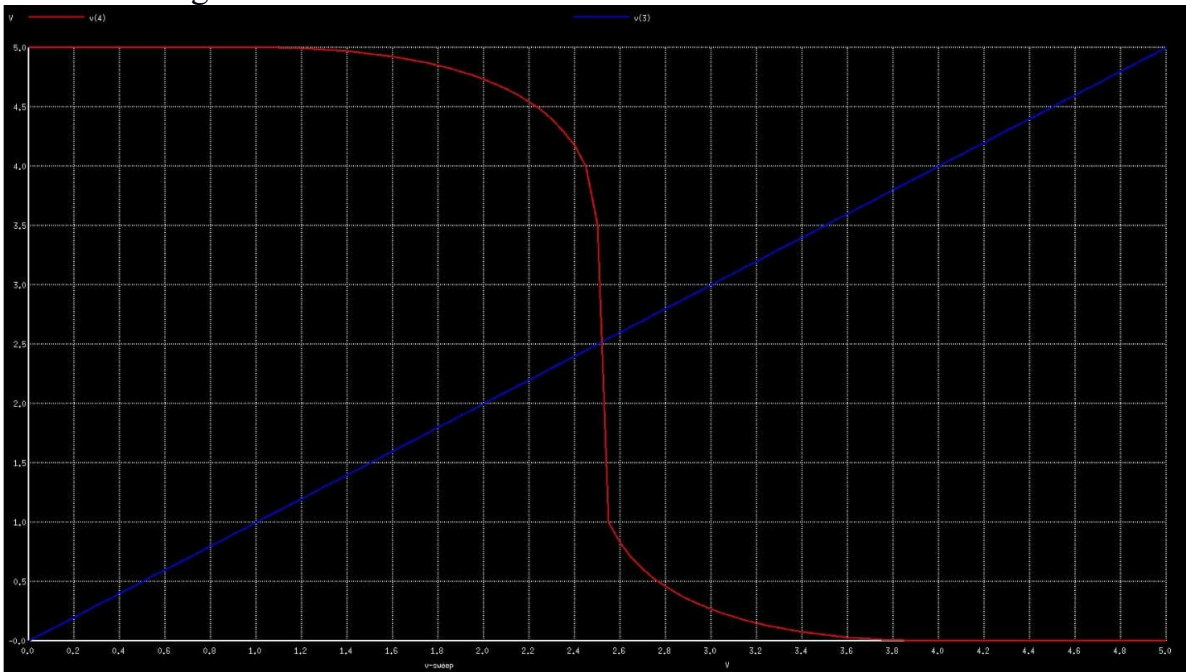
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After Constant Field Scaling (CFS):

```
cmos inverter Neha - 18 (sf = 1.9) cfs
VDD 1 0 DC 2.631V
VIN 3 0 DC 2.631V
VD 1 2 DC 0V
MP 4 3 2 2 PMOD W=10.52U L=0.5263U
MN 4 3 0 0 NMOD W=5.263U L=0.5263U
CL 4 0 2.77pF
.MODEL PMOD PMOS (KP=95U,VTO=-0.5263V)
.MODEL NMOD NMOS (KP=190U,VTO=0.5263V)
.DC VIN 0 5 0.05
.control
run
plot V(4) V(3)
plot deriv(V(4))
plot i(VD)*V(1)
.endc
.end
```

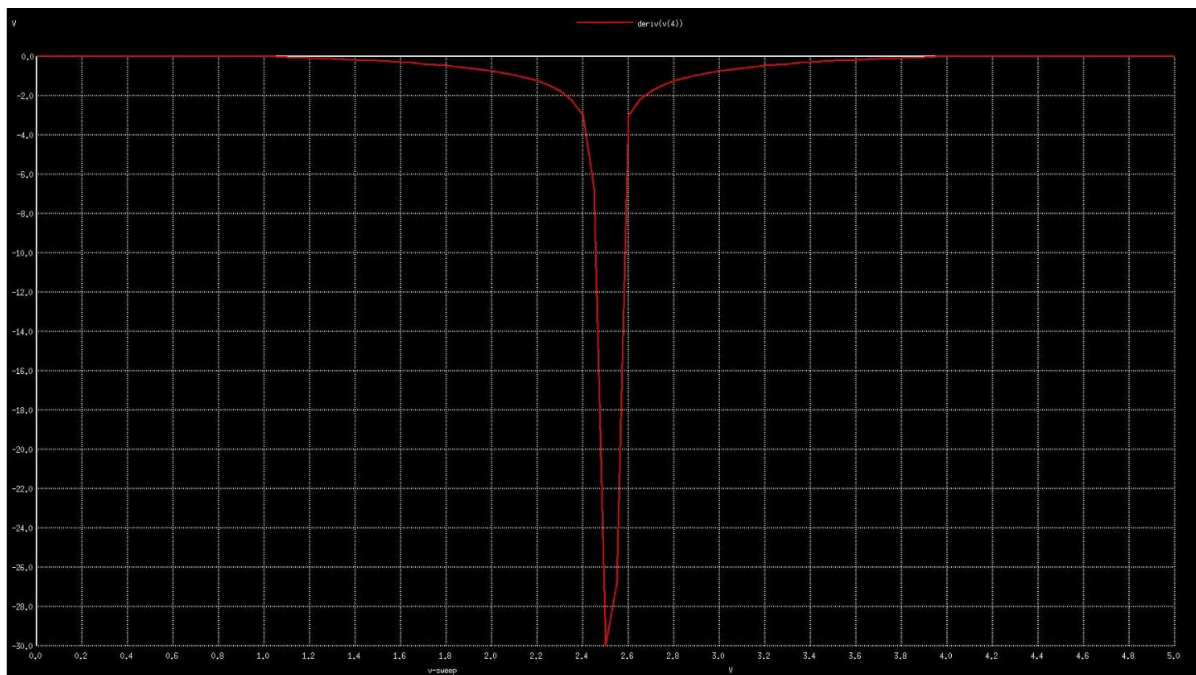
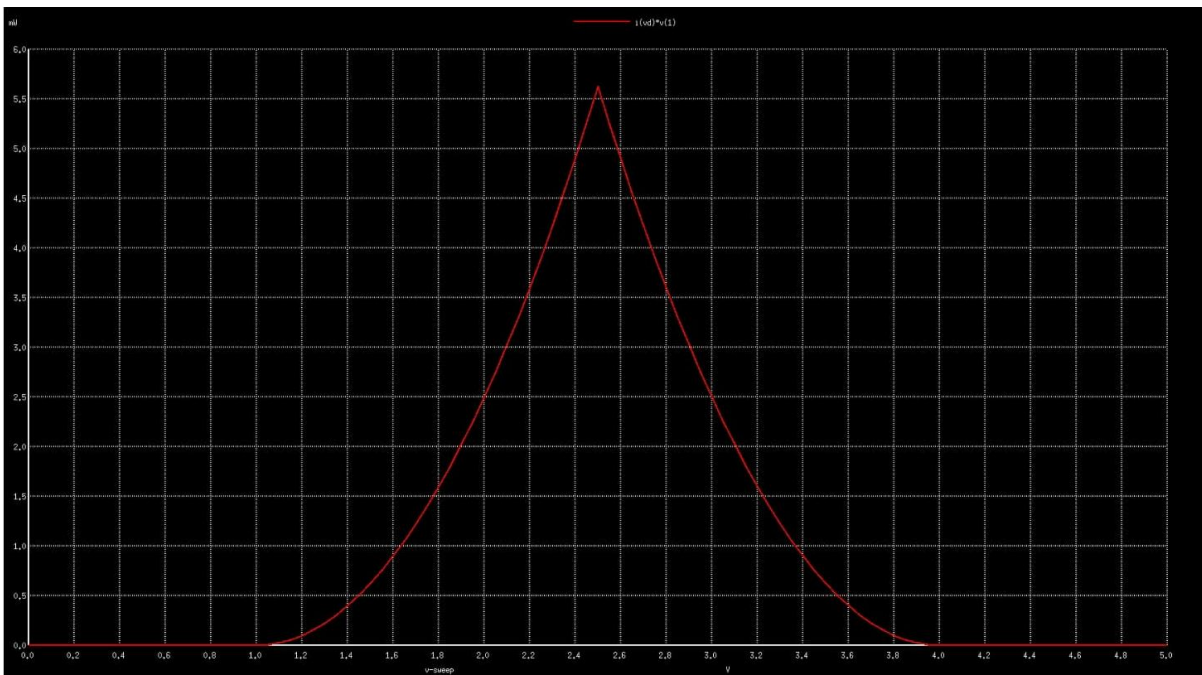
Output:

Before Scaling:





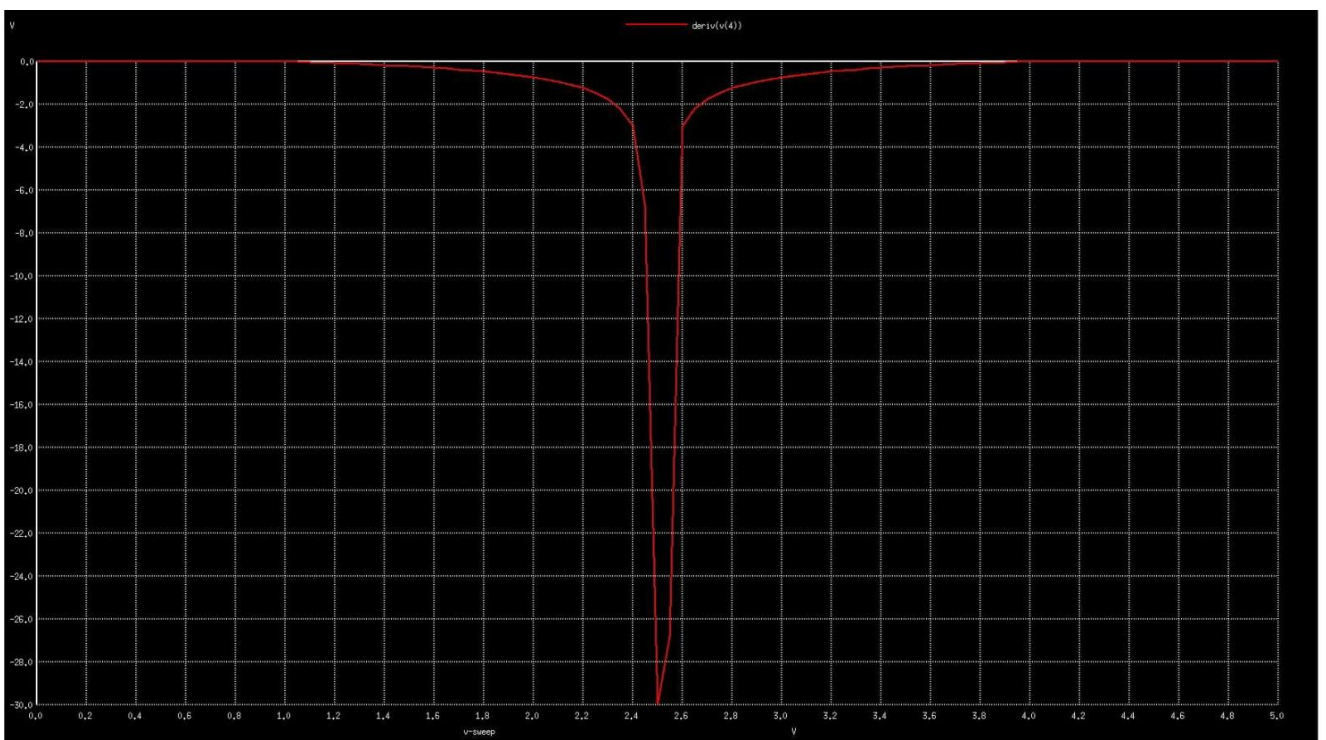
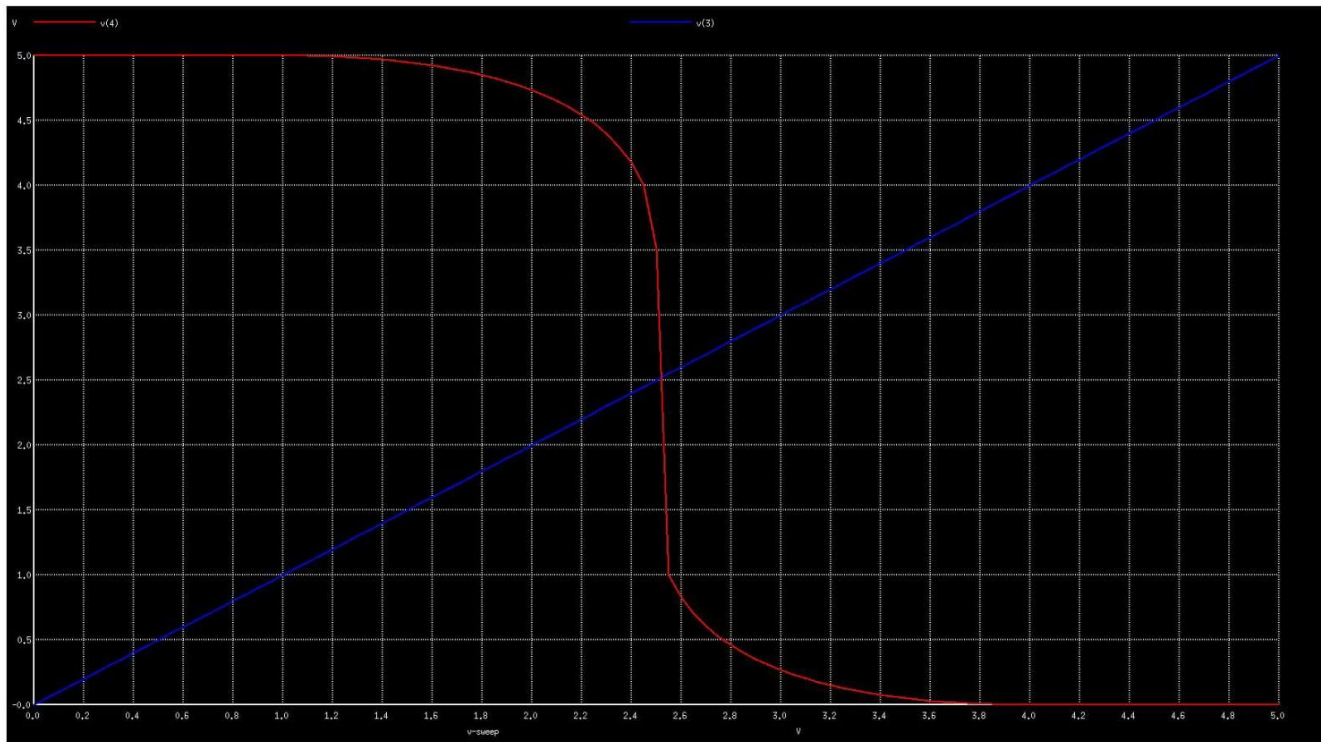
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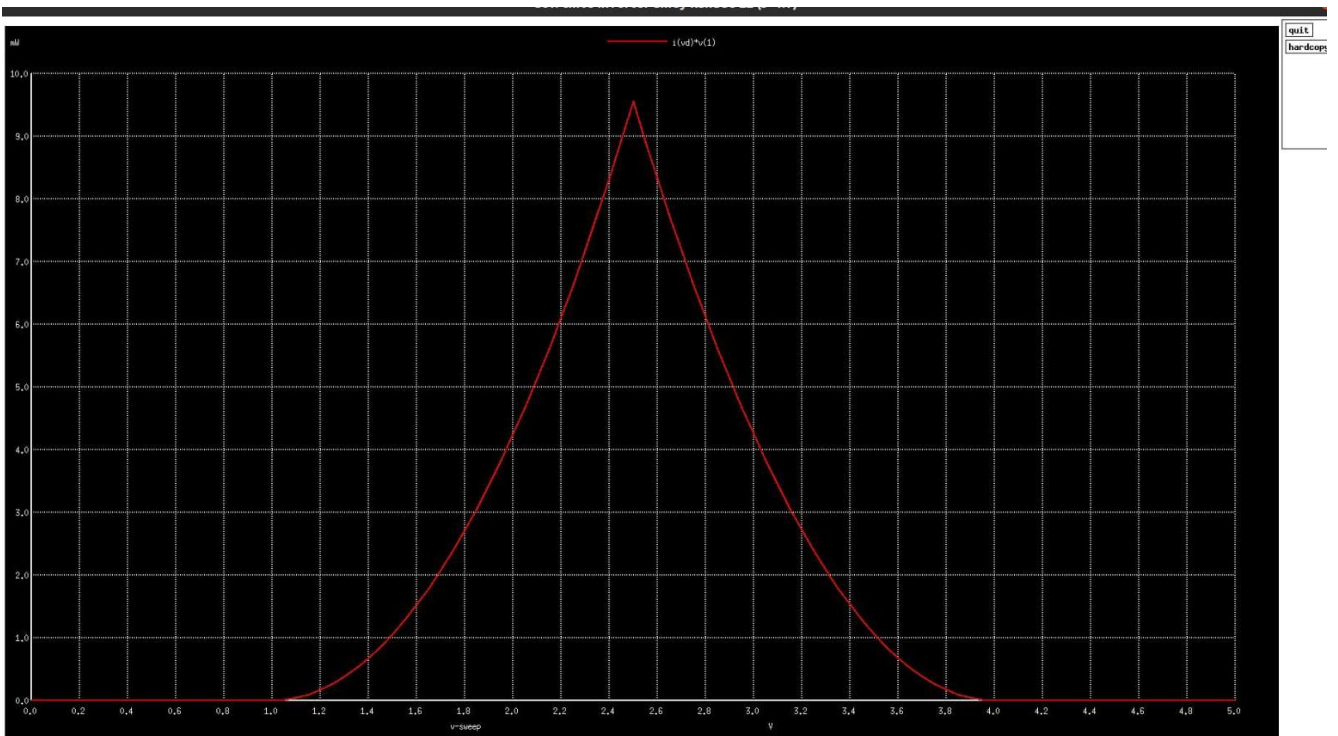
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After Constant Voltage Scaling (CVS):

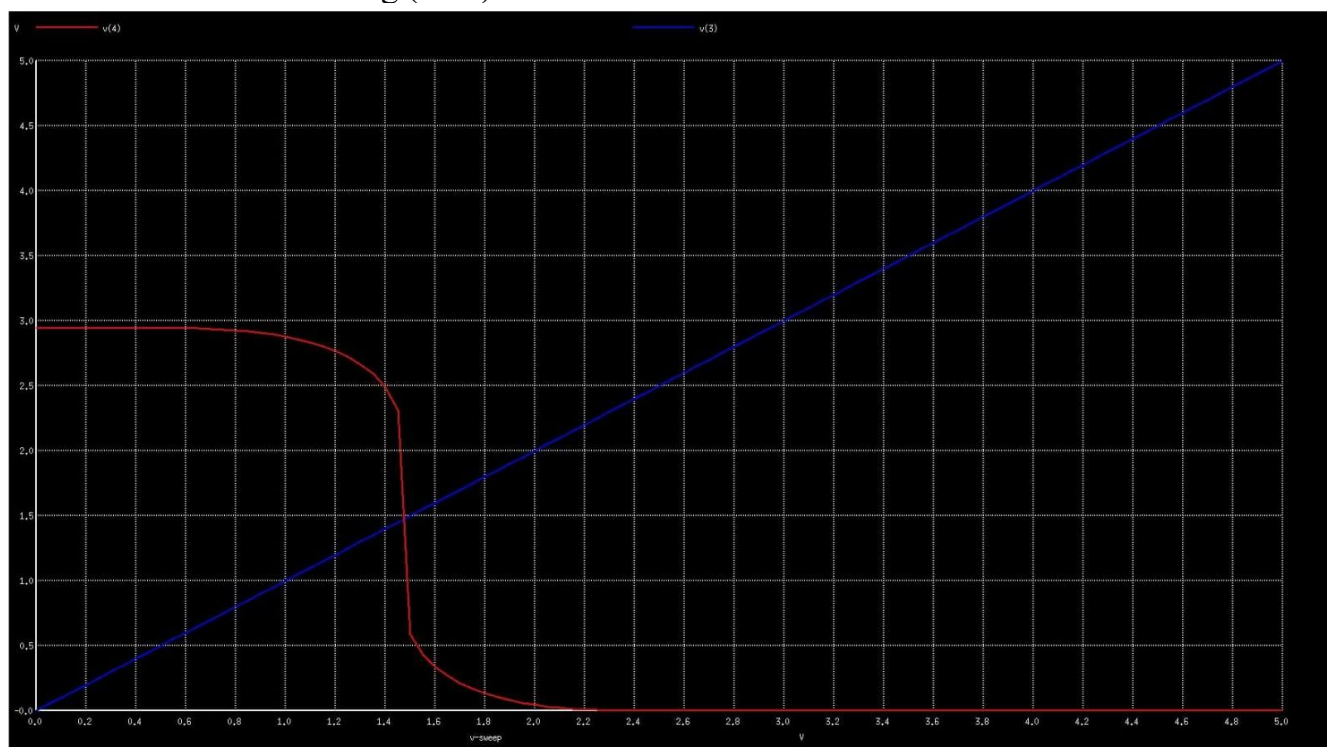




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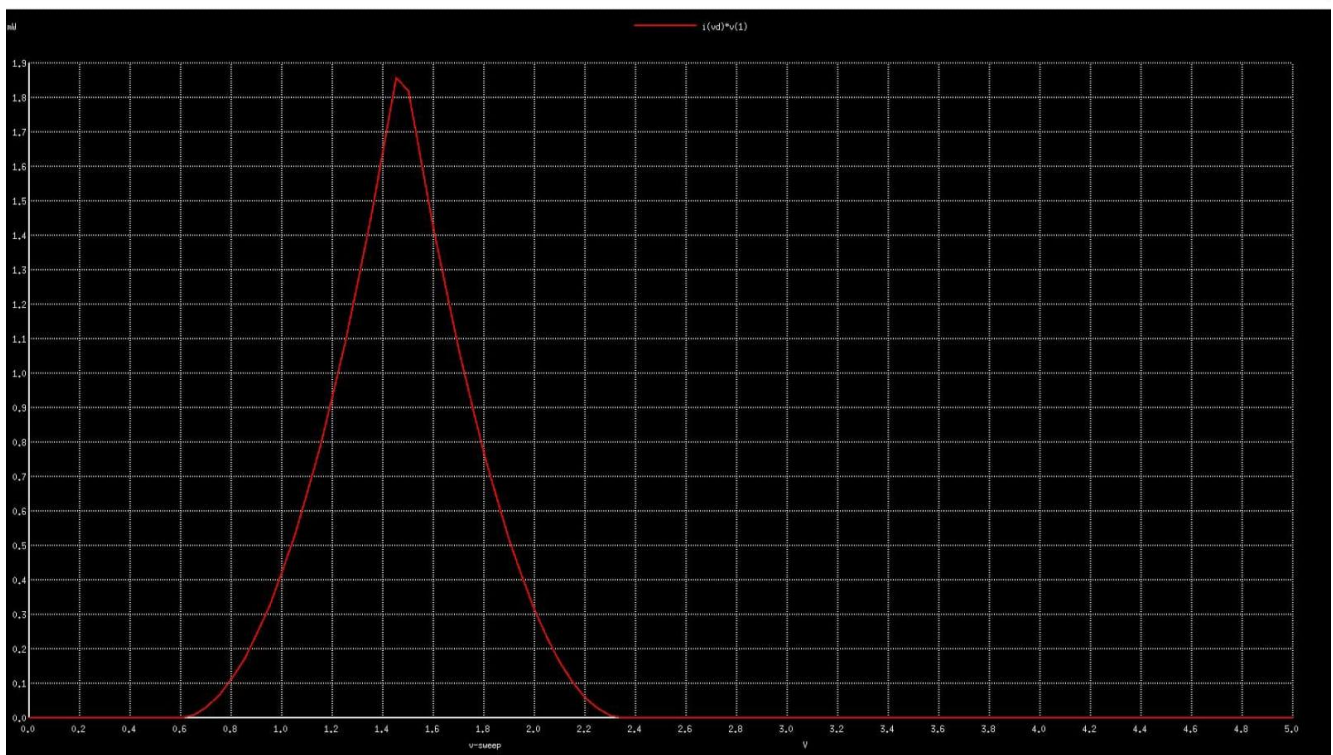
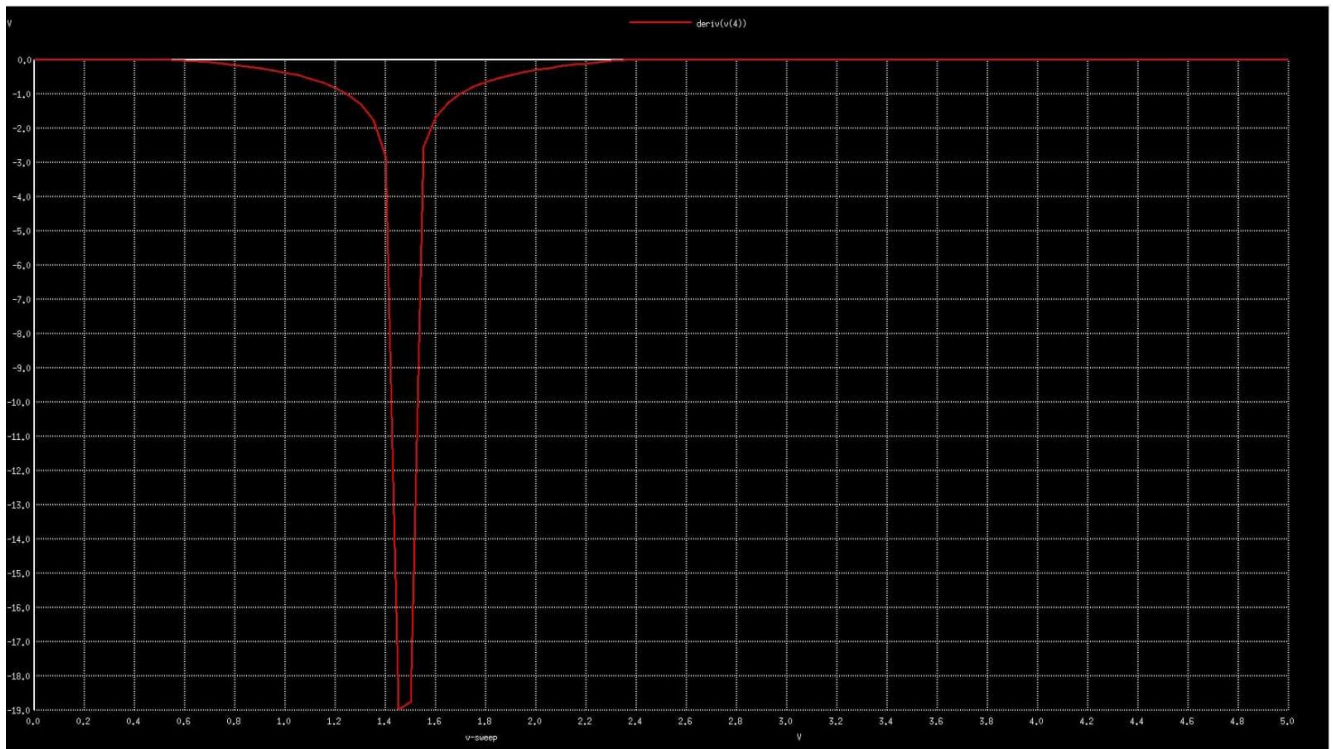


After Constant Field Scaling (CFS):





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Conclusion:
